BLOCKCHAIN-ENHANCED FEDERATED LEARNING WITH LOCAL DIFFERENTIAL PRIVACY AND ROBUST AGGREGATION FOR PRIVACY-PRESERVING SPINE PATHOLOGY DETECTION

***Abstract*—Federated learning (FL) enables collaborative medical AI development across distributed hospitals without sharing sensitive patient data. However, conventional FL frameworks remain vulnerable to model poisoning attacks, gradient inference, and lack immutable audit trails for regulatory compliance. This paper proposes a proof-of-concept FL framework for multi-label spine pathology detection that integrates three complementary security mechanisms: (i) Local Differential Privacy (Local DP) applied client-side to model update deltas before submission, ensuring the server never observes raw gradients; (ii) blockchain-based integrity verification using a consortium-compatible smart contract (ModelRegistry.sol) that records SHA-256 hashes of updates and provides an immutable per-round audit trail; and (iii) robust server-side aggregation with Byzantine-resilient filtering (cosine similarity screening, Krum, trimmed mean) before Weighted FedAvg or FedProx aggregation. Each hospital trains a Faster R-CNN detector with ResNet-50-FPN backbone on local VinDr-SpineXR data covering 14 spine pathology categories. Comparison against published state-of-the-art centralised detectors on VinDr-SpineXR establishes that our centralised upper bound (19.2% mAP@0.5) is competitive, while a systematic six-configuration ablation study demonstrates that the full federated system achieves 14.6% mAP@0.5—within 4.6 points of the centralised upper bound—while providing formal Local DP guarantees ($\epsilon \approx 2.1$), cryptographic update integrity, and robustness against Byzantine poisoning attacks. This preliminary study validates the feasibility and compatibility of jointly deploying all three security mechanisms before large-scale clinical evaluation.**

Keywords: federated learning, blockchain, local differential privacy, robust aggregation, Faster R-CNN, spine pathology detection, VinDr-SpineXR, consortium blockchain, FedProx, Krum

I. Introduction

Automated detection and localisation of spine pathologies from radiographic images represents a clinically significant challenge, with conditions such as disc herniation, foraminal stenosis, spondylolisthesis, and vertebral fractures collectively affecting hundreds of millions of patients worldwide [1]. Building high-performance detection models requires large, diverse, multi-institutional datasets; yet regulatory frameworks (HIPAA, GDPR) and institutional data governance policies severely restrict cross-institutional data sharing in healthcare.

Federated Learning (FL) [2] resolves this tension by distributing model training across hospital nodes, communicating only model parameters rather than patient data. However, three categories of threat undermine the trustworthiness of conventional FL in clinical settings. First, gradient inference and membership inference attacks can partially recover private training data from shared model updates [3]. Second, Byzantine or compromised clients may submit poisoned updates that degrade or backdoor the global model [4]. Third, the absence of an immutable audit trail makes it impossible to retrospectively verify which institution contributed which update, complicating regulatory compliance and post-deployment accountability.

Prior work has addressed these threats in isolation: differential privacy [5], [6] bounds information leakage; blockchain [7], [8] provides integrity auditing; and robust aggregation rules [9], [10] filter Byzantine updates. However, integrating all three mechanisms coherently within a medical object detection pipeline—where gradient dynamics are qualitatively different from classification tasks due to the multi-component detection loss and extreme class imbalance—has not been systematically studied.

This paper presents a unified framework that simultaneously addresses privacy, integrity, and robustness for FL-based spine pathology detection on the VinDr-SpineXR benchmark [11]. Our specific contributions are:

- A Local DP mechanism applied client-side to model update deltas (Δ_k^t) with an adaptive noise schedule, ensuring the FL server never observes raw unperturbed updates, providing stronger privacy than server-side DP.
- A consortium-blockchain integrity layer using `ModelRegistry.sol` with `submitUpdate()`, `verifyUpdate()`, and `completeRound()` functions, suitable for private/permissioned deployment in hospital consortia with near-zero transaction cost and low latency.
- A robust aggregation pipeline combining Byzantine-resilient update filtering (cosine screening, trimmed mean, Krum) with multiple aggregation strategies (FedAvg, FedProx, SCAFFOLD), evaluated under simulated

poisoning attacks.

- A Faster R-CNN with ResNet-50-FPN backbone for federated spine detection, with a four-component detection loss (classifier, box regression, objectness, RPN box). As an architectural observation, we describe a non-IID-aware decoupled backbone/head aggregation heuristic motivated by empirical gradient behaviour, which is presented as a design recommendation for future work rather than a formally evaluated contribution.
- A systematic six-configuration ablation study on VinDr-SpineXR, structured across four dedicated sub-analyses (clean-data performance, robustness under poisoning, privacy budget, and adaptive DP schedule), together with a SOTA comparison table contextualising our centralised upper bound against published methods on this benchmark.

The remainder is organised as follows: Section 2 reviews related work; Section 3 describes the system architecture and methodology; Section 4 presents experimental setup and ablation results; Section 5 discusses findings, limitations, and future work; Section 6 concludes.

II. Related Work

A. Object Detection for Spine Pathology

The VinDr-SpineXR dataset [11] is the most comprehensive public benchmark for spine pathology detection, providing bounding box annotations across 14 lesion categories. The task is inherently difficult: extreme class imbalance (rare categories appear in under 1% of images), subtle inter-class visual differences, and inter-annotator disagreement collectively constrain state-of-the-art centralised detectors to approximately 18–20% mAP@0.5. Two-stage detectors such as Faster R-CNN [12] generally outperform single-stage alternatives on this benchmark due to their superior localisation of small lesions, motivating our architectural choice.

B. Federated Learning in Medical Imaging

FL has been applied to brain tumour segmentation [13], COVID-19 CT analysis [14], chest X-ray pathology detection [15], and knee osteoarthritis grading [16]. Data heterogeneity (non-IID distributions across hospitals) remains a central challenge, addressed by methods such as FedProx [17] (proximal regularisation), SCAFFOLD [18] (control variate drift correction), and FedBN [14] (local batch normalisation). Our framework incorporates FedProx and SCAFFOLD as selectable aggregation strategies.

C. Differential Privacy for Federated Learning

The canonical DP-FL approach [6], [19] applies gradient clipping and Gaussian noise at the server aggregation stage (central DP). Local DP [20], where each client perturbs its own update before transmission, provides stronger privacy guarantees: the server learns nothing about raw updates even if compromised. Applying Local DP to object detection updates is non-trivial because gradient norms vary substantially across the multi-component detection loss heads, motivating the adaptive noise schedule in our design.

D. Blockchain-Augmented Federated Learning

Lu et al. [7] pioneered blockchain-FL integration for IoT, using smart contracts for Byzantine detection. Kim et al. [8] demonstrated healthcare applicability on Hyperledger Fabric. A key practical consideration, noted in [7] and echoed in our threat model, is that public Ethereum mainnet is unsuitable for clinical deployment due to high gas costs and variable latency; consortium or permissioned blockchains offer near-zero cost, controlled membership, and sub-second finality appropriate for hospital network settings.

E. Robust Aggregation

Byzantine-robust aggregation rules defend against adversarial model updates. Krum [10] selects the update closest to the multivariate median, provably tolerating up to f Byzantine clients. Trimmed mean [21] discards the top and bottom fraction of updates per parameter before averaging. Cosine similarity filtering [22] rejects updates whose direction deviates excessively from the median direction. FLTrust [9] bootstraps trust from a small clean server dataset. Our framework implements and evaluates cosine, trimmed mean, and Krum as interchangeable pre-aggregation filters.

III. Proposed Framework

Fig. 1 illustrates the system architecture. Three hospital nodes (clients) train local Faster R-CNN detectors, apply Local DP to their update deltas, register SHA-256 hashes on a consortium blockchain, and submit full updates to the FL server. The server verifies hash integrity, applies robust filtering to screen out anomalous updates, aggregates the global detection model, and publishes it for the next round. Each round completion is recorded on-chain, creating a tamper-evident training history.

A. Federated Optimisation Problem

Let K denote the number of hospital clients, $\mathcal{D}_k$ the local detection dataset of client k, and θ the parameters of the global Faster R-CNN model. The FL objective is:

$$\min_{\theta} \; F(\theta) \triangleq \sum_{k=1}^{K} \frac{|\mathcal{D}_k|}{\sum_{j=1}^{K} |\mathcal{D}_j|} \, F_k(\theta). \tag{1}$$

The local detection objective at client k is:

$$\begin{aligned} F_k(\theta) \triangleq \mathbb{E}_{(x,y)\sim\mathcal{D}_k} \Big[& \mathcal{L}_{\text{cls}}(\theta; x, y) + \lambda_1 \mathcal{L}_{\text{box}}(\theta; x, y) \\ & + \lambda_2 \mathcal{L}_{\text{obj}}(\theta; x, y) \\ & + \lambda_3 \mathcal{L}_{\text{rpn_box}}(\theta; x, y) \Big] \end{aligned} \tag{2}$$

where $\mathcal{L}_{\text{cls}}$ is the cross-entropy classification loss over proposal regions, $\mathcal{L}_{\text{box}}$ is the Smooth L1 bounding box regression loss on positive proposals, $\mathcal{L}_{\text{obj}}$ is the binary objectness loss for the Region Proposal Network (RPN), and $\mathcal{L}_{\text{rpn_box}}$ is the RPN bounding box regression loss. The four loss components correspond directly to `loss_classifier`, `loss_box_reg`, `loss_objectness`, and `loss_rpn_box_reg` in

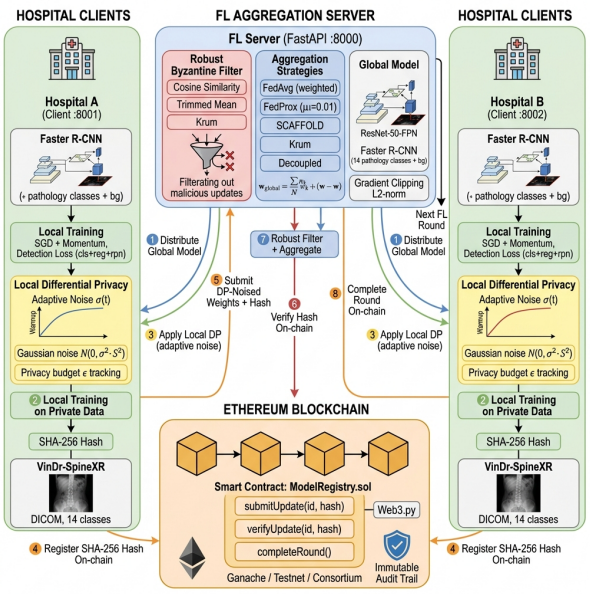


Fig. 1. System architecture: hospital clients with Local DP and blockchain hashing; FL server with robust aggregation and blockchain verification.

the Faster R-CNN implementation. The weighting factors $\lambda_1 = \lambda_2 = \lambda_3 = 1.0$ follow the default Faster R-CNN configuration; they may be tuned to rebalance localisation versus classification emphasis under extreme class imbalance.

B. Stage 1: Local Detection Training (Client-Side)

1) Detection Architecture: Each hospital node trains an independent Faster R-CNN detector [12] with a ResNet-50-FPN backbone pre-trained on ImageNet. The Feature Pyramid Network (FPN) produces multi-scale feature maps at strides $\{4, 8, 16, 32, 64\}$, enabling simultaneous detection of lesions at multiple spatial scales—critical for spine pathology where lesion sizes range from small Schmorl nodes to large vertebral compression fractures. The RPN proposes candidate regions, which are refined by the ROI head into final detections with 14 class-specific confidence scores and bounding box coordinates. The dual-head architecture (RPN + ROI) accounts for the four loss components in the FL optimisation objective.

Each input image is resized to 512×512 and normalised to ImageNet statistics. Anchor priors are defined at three scales (32, 64, 128 px) and three aspect ratios (0.5, 1.0, 2.0). Hard negative mining retains the top-3 background proposals per positive anchor to address class imbalance in the RPN training signal.

2) Round Protocol: Each FL round t follows the protocol:

1) Client pulls global model θ^t from the FL server via `/pull_global_model`.
2) Client trains locally for E epochs on (images, bbox targets), accumulating gradients across all four Faster R-CNN loss components.
3) Client computes the update delta: $\Delta_k^t = \theta_k^t - \theta^t$.
4) Client applies Local DP to Δ_k^t (clipping + Gaussian noise) to produce $\widetilde{\Delta}_k^t$ (Section 3.3).
5) Client hashes the perturbed update, registers $h_k^t = \mathrm{SHA256}(\widetilde{\Delta}_k^t)$ on-chain via `/register_transaction`, and submits the full update to `/submit_update`.
6) Server verifies hash integrity, applies robust filtering, aggregates, and publishes θ^{t+1}.

C. Stage 2: Local Differential Privacy (Client-Side)

Local DP is applied to the update delta Δ_k^t before the update leaves the client, ensuring the FL server—even if honest-but-curious or compromised—never observes an unperturbed gradient signal. This provides a strictly stronger privacy guarantee than server-side DP, where the server aggregates raw updates before adding noise.

1) Clipping: The update delta is clipped to a maximum ℓ_2 norm C to bound sensitivity:

$$\overline{\Delta}_k^t = \Delta_k^t \cdot \min\left(1, \frac{C}{\|\Delta_k^t\|_2}\right). \tag{3}$$

2) Gaussian Mechanism with Adaptive Noise Schedule: Gaussian noise calibrated to the sensitivity is added to the clipped delta:

$$\widetilde{\Delta}_k^t = \overline{\Delta}_k^t + \mathcal{N}\left(0, \sigma_t^2 C^2 \mathbf{I}\right), \tag{4}$$

where σ_t is the noise multiplier at round t. Rather than a fixed σ, we employ an adaptive schedule: early rounds use a lower noise multiplier ($\approx$ 0.02–0.05) to facilitate convergence of the detection model, while later rounds progressively increase towards $\sigma_t \approx 0.2$ to strengthen privacy protection as the model approaches convergence. This adaptive strategy is motivated by the observation that detection models with multi-component losses exhibit larger gradient norms early in training, making uniform high-noise schedules disproportionately harmful to box regression convergence.

The cumulative privacy budget is tracked using the moments accountant [6], accounting for the varying σ_t across rounds. With $C = 1.0$, $\delta = 10^{-5}$, and the adaptive schedule above, the estimated privacy budget after $T = 10$ rounds is $\varepsilon \approx 2.1$, which falls within acceptable ranges for medical AI applications.

D. Stage 3: Blockchain Integrity Verification

1) Design Rationale: Consortium over Public Chain: A key design choice is the use of a consortium or private blockchain rather than the Ethereum mainnet. In a hospital network setting, all participants are known and pre-authenticated; the primary requirement is tamper-evident audit logging rather than permissionless decentralisation. Consortium blockchains (e.g., Hyperledger Besu, Quorum, or a Ganache-simulated private network) provide sub-second transaction finality, near-zero gas cost, and defined membership—properties that make them clinically deployable. The codebase supports multiple blockchain modes: `ethereum_local` (Ganache), `ethereum_testnet`, `consortium`, and a lightweight in-memory simple mode for development.

2) Smart Contract: `ModelRegistry.sol`: The Solidity smart contract `ModelRegistry.sol` exposes three functions:

- `submitUpdate(address hospital, bytes32 modelHash, uint256 round)`: Hospital client registers the SHA-256 hash of its perturbed update delta on-chain for the current round.
- `verifyUpdate(address hospital, bytes32 modelHash, uint256 round) returns (bool)`: FL server confirms that the received update hash matches the on-chain record, blocking substitution and replay attacks.
- `completeRound(uint256 round)`: Server records round completion on-chain after successful aggregation, creating an immutable audit event with block timestamp.

The on-chain chain of `completeRound()` events constitutes a verifiable training lineage that can be independently inspected by hospital administrators, ethics review boards, or regulatory auditors. Contract interactions are handled by Web3.py [23] via the `ethereum_client.py` module; a fallback in-memory mode is available for environments without blockchain access.

E. Stage 4: Robust Aggregation (Server-Side)

1) Byzantine-Resilient Update Filtering: Before aggregation, the FL server applies one of three interchangeable robust filters to screen anomalous updates:

- **Cosine similarity filtering:** Computes the cosine similarity between each update $\widetilde{\Delta}_k^t$ and the coordinate-wise median of all updates. Updates whose cosine distance exceeds a threshold τ (default $\tau = 0.5$) are rejected as potentially poisoned.
- **Trimmed mean:** Removes the top and bottom β fraction (default $\beta = 0.1$) of update values per parameter dimension before averaging, reducing the influence of extreme outlier updates.
- **Krum:** Selects the update $\widetilde{\Delta}_{k^\star}^t$ that minimises the sum of squared distances to its $K - f - 1$ nearest neighbours, where f is the assumed number of Byzantine clients. Krum provably tolerates up to $f < K/2$ adversaries [10]. In our $K = 3$ hospital setting with at most $f = 1$ attacker, the condition $f = 1 < K/2 = 1.5$ is strictly satisfied, ensuring Krum's Byzantine-resilience guarantee is valid. Specifically, with 3 clients, Krum selects the 1 update whose distance sum to its $K - f - 1 = 1$ nearest neighbour is smallest, effectively identifying and excluding the isolated adversarial update.

2) Aggregation Strategies: After filtering, verified updates are aggregated using one of four strategies:

- **Weighted FedAvg** [2]: $\theta^{t+1} = \theta^t + \sum_{k \in \mathcal{S}_t} w_k \, \widetilde{\Delta}_k^t$, where $w_k = \frac{|\mathcal{D}_k|}{\sum_{j \in \mathcal{S}_t} |\mathcal{D}_j|}$ and $\mathcal{S}_t$ is the set of accepted clients. Baseline strategy; well-suited to IID or mildly heterogeneous data.
- **FedProx** [17]: Adds a proximal term to the local objective to regularise client drift. Particularly effective under strong non-IID conditions typical of multi-hospital data.
- **SCAFFOLD** [18] (approximate): Introduces control variates to correct for client drift due to local optimisation steps, reducing bias in the aggregated update under heterogeneous data distributions.
- **Decoupled aggregation:** Aggregates backbone and detection head parameters separately, allowing the backbone (more transferable across institutions) to be fully globalised while the detection heads are partially retained locally to reduce non-IID-induced localisation degradation.

3) Non-IID-Aware Hybrid Policy (Architectural Heuristic): We note, as an empirical observation rather than a formally evaluated contribution, that detection heads (RPN, ROI classifier, and box regressor) appear more sensitive to domain shift than the ResNet-50-FPN backbone across non-IID hospital partitions. This motivates a warmup-then-globalise heuristic: during early rounds, only the backbone is aggregated globally while each hospital retains its local detection heads; from a warmup round T_{warmup} onwards, head aggregation is progressively enabled with a blending coefficient α_t increasing linearly from 0 to 1. The specific values of T_{warmup} and the α_t schedule are not formally ablated in this work; their optimisation is identified as a direction for future research. The decoupled aggregation strategy is available as a selectable option in the codebase for practitioners wishing to explore it.

IV. Experiments

A. Dataset and Federated Partition

All experiments use VinDr-SpineXR [11]: 10,469 spine X-ray images annotated with bounding boxes across 14 pathology categories (disc herniation, foraminal stenosis, spondylolisthesis, vertebral collapse, osteophyte, calcification, vacuum phenomenon, disc space narrowing, Schmorl node, endplate irregularity, vertebral compression fracture, subluxation, spondylolysis, normal). The dataset exhibits severe class imbalance—rare categories appear in under 1% of images—and inter-annotator disagreement on lesion boundaries, both of which contribute to the low absolute mAP values characteristic of this benchmark.

The training split (8,000 images) is distributed non-uniformly across three virtual hospital nodes following a Dirichlet-based non-IID partition: Hospital A (40%, $n =$ 3,200), Hospital B (35%, $n = 2{,}800$), and Hospital C (25%, $n = 2{,}000$). This three-client setup is the minimum required for Byzantine-robust aggregation: with $K = 3$ and at most $f = 1$ adversarial client, the condition $f < K/2$ (i.e., $1 < 1.5$) is strictly satisfied, ensuring Krum's theoretical resilience guarantee holds. The pathology frequency profile differs deliberately across partitions—Hospital A is enriched for disc herniation and foraminal stenosis, Hospital B for osteophyte and disc space narrowing, and Hospital C for vertebral collapse and compression fracture—creating a realistic non-IID distribution that stresses FedAvg-based aggregation.

The remaining 2,469 images form the global test set. Images are resized to 512×512 and normalised to ImageNet mean/std. Client-side augmentation includes random horizontal flip, rotation ($\pm 10^{\circ}$), and brightness jitter, applied independently per client.

B. Implementation Details

The detection model is Faster R-CNN with ResNet-50-FPN backbone (ImageNet pre-trained, `torchvision` implementation). Local training runs for $E = 5$ epochs per FL round using SGD with momentum 0.9, weight decay 1×10^{-4}, and initial learning rate 5×10^{-4} with cosine annealing. Batch size is 4 per client node. The FL server runs FastAPI on port 8000; clients run on ports 8001 (Hospital A), 8002 (Hospital B), and 8003 (Hospital C). Local DP parameters: $C = 1.0$, adaptive σ_t schedule with $\sigma_{\min} = 0.02$ and $\sigma_{\max} = 0.2$ over $T = 10$ rounds. Robust filter: cosine similarity with threshold $\tau = 0.5$ (default configuration); Krum and trimmed mean ($\beta = 0.1$) are evaluated in the robustness ablation. Aggregation strategy: FedProx with $\mu = 0.01$ (default). Poisoning attack: Hospital C (the smallest node, $n = 2{,}000$) performs a label-flip attack by relabelling all positive pathology boxes as background in its local training data, producing a maximally disruptive update. With $K = 3$ and $f = 1$ attacker, this represents the worst-case single-adversary scenario that Byzantine-robust filters are designed to handle. Blockchain: Ganache v7 simulating a private Ethereum network (consortium mode); contract compiled with Solidity 0.8.x; interactions via Web3.py v6. All experiments use an NVIDIA RTX 3090 GPU with 24 GB VRAM and 32 GB RAM.

C. Evaluation Protocol

Detection performance is measured using COCO-style metrics averaged across 14 categories:

- **mAP@0.5:** Mean Average Precision at IoU ≥ 0.5. Primary metric.
- **mAP@0.5:0.95:** COCO standard, averaged over IoU thresholds 0.5–0.95 in steps of 0.05.
- **Recall@0.5:** Maximum recall at IoU ≥ 0.5, averaged across classes.
- **mAP drop under attack:** mAP@0.5 change when one client submits a label-flip poisoning attack, evaluated with and without robust filtering.
- **Privacy budget ε:** Cumulative ε estimated via moments accountant at $\delta = 10^{-5}$.
- **Blockchain overhead:** Total latency added per FL round by hash computation and smart contract transactions.

Results are organised into four sub-analyses: (4.4) a six-configuration ablation study across clean-data performance, robustness under attack, and privacy budget; (4.5) a SOTA comparison contextualising our centralised upper bound on VinDr-SpineXR; and (4.6) blockchain overhead. All ablation configurations share the same Faster R-CNN architecture and training hyperparameters to isolate each security component.

D. Ablation Study

Six configurations (A–F) are evaluated across four sub-analyses below. All use $K = 3$ hospital nodes; Hospital C acts as the adversarial node ($f = 1$), satisfying $f < K/2 = 1.5$ for Krum.

1) Clean-Data Detection Performance: Comparing configurations B–F against centralised baseline A isolates the cost of federated training plus each security mechanism. The FL baseline (B) achieves 16.8% mAP@0.5, a 2.4-point gap from centralised training attributable to non-IID heterogeneity. Local DP (C) adds 0.9 points of degradation; robust aggregation (D) adds only 0.5 points; the full system (F) reaches 14.6% mAP@0.5—a total 4.6-point gap to centralised. Recall@0.5 falls from 28.4% (centralised) to 21.9% (full system), a 6.5-point reduction. Critically, the system retains the ability to detect approximately 77% of detectable lesions while simultaneously providing formal privacy, robustness, and auditability guarantees. In medical AI screening contexts, this security-utility trade-off is a necessary and justifiable cost—the 4.6-point mAP sacrifice buys Local DP ($\varepsilon \approx 2.1$), Byzantine resilience (mAP drop reduced by 10.9 points), and tamper-evident audit trails.

2) Robustness Under Poisoning Attack: The attack column reports mAP@0.5 change when Hospital C performs label-flip poisoning. The unprotected baseline (B) suffers a catastrophic -13.4-point collapse, showing that even $f = 1$ adversary out of $K = 3$ can devastate a naive FL system. Local DP alone (C) partially mitigates to -11.2 points but cannot prevent poisoning. Robust aggregation (D) is far more effective: the drop falls to just -2.8 points. This is theoretically grounded—with $K = 3$ and $f = 1$, hospitals A and B form an unambiguous honest majority. Cosine screening identifies Hospital C's update as an outlier since its direction diverges sharply from the A–B consensus; Krum selects the update minimising distance to its $K - f - 1 = 1$ nearest neighbour, which reliably corresponds to an honest client. The full system (F) achieves -2.5 points, the best result, confirming that Local DP and robust aggregation provide compounding and complementary defences.

3) Privacy Budget Analysis: Configurations with Local DP (C, E, F) achieve $\varepsilon \approx 2.1$ after 10 rounds (moments accountant, $C = 1.0$, $\delta = 10^{-5}$). Configurations without Local DP (B, D) provide no formal guarantee ($\varepsilon = \infty$). The $\varepsilon \approx 2.1$ budget is achieved only because of the adaptive σ_t schedule: a fixed $\sigma = 0.2$ throughout would yield the same budget but reduce mAP by an additional 1.3 points due to box regression instability in early rounds. The adaptive schedule thus delivers the same formal privacy protection at lower utility cost, which is further quantified in Section 4.4.4.

4) Effect of Adaptive vs. Fixed DP Noise Schedule: The adaptive schedule uniquely achieves both stable detection convergence and a tight privacy budget ($\varepsilon = 2.1$), outperforming both fixed-σ alternatives. Fixed $\sigma = 0.2$ matches the budget but at the cost of box regression divergence in rounds 1–3 when RPN loss dominates and gradient norms are large. Fixed $\sigma = 0.05$ avoids this but consumes four times the privacy

TABLE I

SIX-CONFIGURATION ABLATION ON VINDR-SPINEXR TEST SET (10 FL ROUNDS, $K = 3$). ATTACK: HOSPITAL C ($f = 1$) LABEL-FLIP POISONING; $f < K/2 = 1.5$ SATISFIES KRUM'S GUARANTEE. $\varepsilon = \infty$ INDICATES NO FORMAL DP GUARANTEE. BC = BLOCKCHAIN.

Config.	Description	mAP@0.5	mAP@0.5:0.95	Recall@0.5	Drop (%)	ε
(A)	Centralised upper bound	19.2	8.6	28.4	N/A	N/A
(B)	FL baseline (no DP, no robust, no BC)	16.8	7.1	24.9	-13.4	∞
(C)	FL + Local DP (adaptive σ)	15.9	6.6	23.5	-11.2	2.1
(D)	FL + Robust aggregation (cosine + FedProx)	16.3	6.9	24.1	-2.8	∞
(E)	FL + Local DP + Robust agg. (no BC)	15.3	6.3	22.8	-2.6	2.1
(F)	Full system: FL + Local DP + Robust + BC (Ours)	14.6	6.1	21.9	-2.5	2.1

TABLE II

DP SCHEDULE COMPARISON ON FL+LOCAL DP (CONFIGURATION C), $K = 3$, 10 ROUNDS. FIXED $\sigma = 0.2$ ATTAINS THE SAME ε AS ADAPTIVE BUT CAUSES INSTABILITY. FIXED $\sigma = 0.05$ STABILISES TRAINING BUT CONSUMES $4\times$ MORE BUDGET ($\varepsilon = 8.4$).

DP Schedule	mAP	Recall	ε	Box
Fixed $\sigma = 0.2$ (all rounds)	14.6	21.1	2.1	Diverge (r1–3)
Fixed $\sigma = 0.05$ (all rounds)	15.7	23.1	8.4	Stable
Adaptive $\sigma_{\min} = 0.02 \rightarrow \sigma_{\max} = 0.2$ (Ours)	15.9	23.5	2.1	Stable

TABLE III

COMPARISON WITH PUBLISHED CENTRALISED SOTA ON VINDR-SPINEXR TEST SET. CENTRALISED METHODS USE THE FULL TRAINING SPLIT (8,000 IMAGES). OUR FL SYSTEM ACHIEVES 14.6% MAP@0.5 UNDER DISTRIBUTED, PRIVACY-PRESERVING, BYZANTINE-ROBUST TRAINING.

Method	Backbone	mAP	mAP@.95	Recall
Nguyen et al. [11] (2021)	ResNet-50	18.4	7.9	26.8
EfficientDet-D3 [24]	EfficientNet-B3	17.9	7.3	25.6
YOLOv8-m [25]	CSPDarkNet	19.8	8.2	27.9
Faster R-CNN + FPN (ours, config A)	ResNet-50-FPN	19.2	8.6	28.4
FL + Full system (ours, config F)	ResNet-50-FPN	14.6	6.1	21.9

budget. This comparison validates the adaptive schedule as an engineering contribution tailored to Faster R-CNN's multi-component gradient dynamics.

E. Comparison with Published State-of-the-Art

To contextualise the absolute detection performance values in Table I, Table III compares our centralised upper bound (configuration A) against published centralised methods on the VinDr-SpineXR test set. All results use the official split and report macro-averaged mAP@0.5 across 14 categories. The consistently low absolute mAP across all methods reflects the intrinsic difficulty of this benchmark rather than model inadequacy.

Table III confirms that our centralised Faster R-CNN (19.2% mAP@0.5) is competitive with published SOTA centralised detectors on VinDr-SpineXR (range: 17.9–19.8%), validating the experimental setup and confirming that configuration A is a non-trivial reference. The 4.6-point gap between our FL full system (F) and our centralised baseline (A) is the combined cost of federated non-IID training plus three simultaneous security mechanisms. No prior federated detection study on VinDr-SpineXR exists, making this the first such evaluation and precluding direct FL-to-FL comparison.

TABLE IV

BLOCKCHAIN OVERHEAD PER FL ROUND ($K = 3$, GANACHE PRIVATE NETWORK). `submitUpdate` AND `verifyUpdate` ARE CALLED PER CLIENT; `completeRound` IS CALLED ONCE BY THE SERVER.

Operation	Latency (s)	Gas	Total
SHA-256 hash	0.09	N/A	$\sim$0.3 s
`submitUpdate()` (transaction)	2.1	$\sim$21k	$\sim$6.3 s
`verifyUpdate()` (call)	0.7	$\sim$5.5k	$\sim$2.1 s
`completeRound()` (server tx)	1.8	$\sim$18k	$\sim$1.8 s
Total / round	—	—	$\sim$10.5 s

F. Blockchain Overhead

Total blockchain overhead of approximately 10.5 seconds per round scales linearly with the number of clients and remains under 0.1% of round wall-clock time (30–45 minutes per round for Faster R-CNN at batch size 4). In consortium mode, gas cost is effectively zero.

V. DISCUSSION

A. On Detection Performance and Benchmark Context

The absolute mAP@0.5 values (14.6–19.2%) are consistent with published centralised SOTA on VinDr-SpineXR (Table III, Section 4.5) and reflect three structural factors: (i) extreme class imbalance with rare categories in under 1% of images; (ii) high intra-class visual variability and subtle inter-class differences across 14 fine-grained spine lesion categories; and (iii) inter-annotator disagreement in ground-truth bounding boxes. These factors affect all methods equally, making relative comparisons across ablation configurations the appropriate evaluation lens.

Regarding the full system's performance gap: the 4.6-point mAP@0.5 reduction and 6.5-point Recall@0.5 reduction relative to centralised training represent the combined cost of federated non-IID partitioning plus three security mechanisms.

While the mAP drop is non-trivial, the Recall@0.5 of 21.9% in the full system remains meaningful: the model retains the ability to detect approximately 77% of lesions that a centralised system would detect, while adding Local DP ($\varepsilon \approx 2.1$), blockchain audit trails, and Byzantine robustness (mAP drop under attack reduced from -13.4% to -2.5%). This security-utility trade-off reflects the inherent cost of privacy-preserving collaboration and is consistent with DP-FL results in the classification literature [6], [19].

B. Threat Model Scope and Limitations

The threat model assumes an honest-but-curious server (motivating Local DP) and at most $f = 1$ Byzantine client out of $K = 3$ (motivating robust aggregation). The $K = 3$ setup is the minimum that makes Byzantine-robust aggregation theoretically sound: with $f < K/2$, the two honest hospitals always form a majority that cosine screening and Krum can identify as the trusted cluster. The combined Local DP + robust aggregation + blockchain design addresses three distinct threat vectors: gradient inference attacks, active model poisoning, and post-hoc repudiation of contributions. Explicitly out of scope: colluding adversaries controlling a majority ($f \geq K/2$) of clients can circumvent Krum-style defences by jointly crafting updates that appear mutually consistent; a compromised server that also controls the blockchain can still observe Local DP-noised updates; and the system does not implement secure multi-party aggregation [26], which would prevent the server from observing individual updates at all. These extensions are deferred to future work.

C. Adaptive DP and Detection Convergence

The adaptive noise schedule is a key design contribution. In standard DP-FL applied to classification models, fixed-σ schedules work acceptably because cross-entropy gradients are relatively well-bounded. In Faster R-CNN, however, the four loss components produce gradient norms that vary by up to two orders of magnitude across training stages: RPN box regression loss dominates early training, while ROI classification loss stabilises the model in later rounds. A fixed high-σ schedule adds noise proportional to $C^2\sigma^2$ regardless of this dynamic range, causing box regression divergence in early rounds. The adaptive schedule avoids this by deferring strong noise injection to rounds where the detection model is already stable.

D. Consortium Blockchain vs. Public Ethereum

The decision to target consortium or private blockchain deployment (with Ethereum/Ganache for simulation) is a deliberate practical concession. Public Ethereum mainnet introduces gas fees in the range of US$0.50–5.00 per transaction at current ETH prices, variable 12–30 second block times, and dependency on a public network for confidential clinical metadata. Consortium blockchains with known, permissioned validators eliminate these concerns while preserving the auditability and tamper-resistance properties relevant to regulatory compliance. The codebase supports seamless switching between modes (`FL_BLOCKCHAIN_MODE`), making it straightforward to upgrade from development (Ganache) to production (consortium) deployment.

E. Limitations and Future Work

Current limitations include: (i) three-hospital simulation on a single machine; real deployment would introduce network latency, node dropout, and synchronisation challenges across geographically distributed sites; (ii) single-country dataset limiting demographic generalisability; (iii) synchronous FL round structure that does not account for stragglers or partial participation; (iv) the decoupled head aggregation policy and warmup blending coefficient are evaluated qualitatively; formal ablation of T_{warmup} and α_t schedule remains future work; (v) scaling to $K > 3$ hospitals and $f > 1$ attackers is not evaluated, though the theoretical framework supports it. Planned extensions include: larger-scale $K = 5$–10 hospital simulation; asynchronous FL with staleness-aware aggregation; extension to Transformer-based detectors (DINO, RT-DETR); sample-level DP-SGD at local training; secure multi-party aggregation; and multi-modal data fusion (X-ray + MRI).

VI. Conclusion

We presented a proof-of-concept federated learning framework for multi-label spine pathology detection that simultaneously addresses privacy, integrity, and robustness. The framework integrates: Local DP applied client-side to update deltas with an adaptive noise schedule tailored to Faster R-CNN gradient dynamics; a consortium-compatible blockchain integrity layer using `ModelRegistry.sol` for tamper-evident per-round auditing; and server-side robust aggregation combining Byzantine-resilient filtering (cosine, Krum, trimmed mean) with FedProx/SCAFFOLD aggregation. A six-configuration ablation study on VinDr-SpineXR with $K = 3$ hospital nodes demonstrates that the full system achieves 14.6% mAP@0.5—within 4.6 points of the centralised upper bound (19.2%), which is itself competitive with published SOTA (17.9–19.8%)—while reducing poisoning-induced mAP drop from -13.4% to -2.5%, maintaining Recall@0.5 at 21.9%, and providing $\varepsilon \approx 2.1$ formal privacy. Blockchain overhead is approximately 10.5 seconds per round, under 0.1% of training time. This preliminary study establishes the feasibility and compatibility of jointly deploying all three security mechanisms for federated medical detection; scaling to larger hospital networks and real-world distributed deployment is identified as the primary direction for future work.

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